

A STUDY OF RECOVERING QUAD-POLARIMETRIC INFORMATION FROM SLC/MLC COMPACT-POLARIMETRIC SAR DATA VIA CNNs

Xinling Du¹, Weidong Sun¹, Jie Yang¹, Lingli Zhao², Lei Shi¹

¹ State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, China

² School of Remote Sensing and Information Engineering, Wuhan University, China

ABSTRACT

Quad polarimetric (quad-pol) Synthetic Aperture Radar (SAR) has unique advantages to record the detailed backscattering information, which significantly enhances the interpretation quality of objects. However, the quad-pol SAR system's intricate nature imposes constraints on its swath coverage and spatial resolution. A promising solution to this challenge lies in the recovery of quad-pol information from Compact-Polarimetric (CP) SAR data. In this paper, recovery methods based on the convolutional neural network (CNN) are introduced, for both single-look complex (SLC) and multi-look complex (MLC) SAR data. It also presents a comparative performance analysis of these recovery techniques across two data types, emphasizing the feasibility and efficiency of two approaches in overcoming the limitations of quad-pol SAR systems while maintaining high data quality.

Index Terms—Compact Polarimetry; CNN; Synthetic Aperture Radar; Quad Polarimetry.

1. INTRODUCTION

With the technical advancement from single/dual polarization to quad-pol observation modes, the target backscattering information obtained by SAR is gradually increasing. However, this is accompanied by increasing requirements on radar hardware systems, making it difficult to avoid the contradictions among scattering information, spatial details, and observation bandwidth.

CP necessitates fewer hardware and software resources compared to conventional quad-pol SAR systems. It offers high-resolution imaging and captures more comprehensive ground target information than single or dual-polarization systems. Thus, CP SAR effectively balances hardware resource requirements with the acquisition of detailed scattering information^{[1][2][3][4]}.

The first approach to leveraging the quad-pol information in CP SAR involves direct feature extraction from CP, while a secondary approach, discussed herein, seeks to maximize information retrieval. This enables the application of established quad-pol data processing and feature extraction techniques^[5]. Historically, several strategies have been

developed to reconstruct quad-pol data from compact polarimetric SAR data. In 2005, Jean-Claude Souyris^[6] introduced three methodologies reliant on reflection symmetry, rotational symmetry, and the amalgamation of both to restore quad-pol data from CP, contingent on specific prior assumptions.

Considering the complexity of electromagnetic wave interaction with ground targets, restoring quad-pol information from CP is a complex and abstract nonlinear mapping process. CNN used for restoring compact polarized SAR remote sensing image data exhibit good generalization capability and scalability. In 2021^[7], a method was proposed for restoring quad-pol covariance matrix C_3 from CP covariance matrix C_2 , this method employ prior assumptions, primarily targeting MLC SAR data for the recovery of C_3 and T_3 matrices, which can reduce the resolution of remote sensing images and damage spatial details, rendering them unsuitable for tasks requiring high resolution.

Therefore, this paper employs a deep learning-based approach for quad-pol information recovery. This method not only allows for the restoration of quad-pol images on MLC SAR data, but also enables the restoration of quad-pol scattering matrix S_2 on SLC SAR data without introducing prior assumptions. Furthermore, this paper compares the effectiveness of two techniques for restoring quad-pol information and evaluates the accuracy of the restoration results.

2. METHODOLOGY

Up to now, three CP modes have been proposed: the $\pi/4$ mode^[8], Dual Circular Polarimetric (DCP) mode^[9], and Hybrid Polarimetric (HP) mode. This paper demonstrates the effectiveness of quad-pol recovery using the $\pi/4$ mode as an example. In this mode, a 45° linearly polarized wave is transmitted, and horizontal (H) and vertical (V) linearly polarized waves are received.

The target vector Ω for CP is (1). The symbol S denotes the complex backscattering element.

$$\begin{aligned}\Omega_{CP} &= \begin{bmatrix} S_{\pi/4}^H & S_{\pi/4}^V \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} & S_{VH} \\ S_{HV} & S_{VV} \end{bmatrix} \begin{bmatrix} 1 \\ \pm 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} \pm S_{HV} \\ S_{VH} \pm S_{VV} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} \pm S_{VH} & S_{HV} \pm S_{VV} \end{bmatrix} \end{aligned} \quad (1)$$

The polarimetric covariance matrices for the $\pi/4$ mode CP data are given by (2). The variables J represent the elements of the 2×2 CP covariance matrix when operating in the $\pi/4$ mode. The angle brackets $\langle \cdot \rangle$ signify ensemble averaging, and the superscript asterisk $*$ indicates complex conjugation. The real and imaginary parts of a complex number are denoted by $\text{Re}(\cdot)$ and $\text{Im}(\cdot)$.

$$C_{CP} = \begin{bmatrix} J_{11} & J_{12} \\ J_{12}^* & J_{22} \end{bmatrix} \quad (2)$$

$$= \frac{1}{2} \left[\begin{array}{cc} \langle |S_{HH}|^2 \rangle & \langle S_{HH} S_{VV}^* \rangle \\ \langle S_{VV} S_{HH}^* \rangle & \langle |S_{VV}|^2 \rangle \end{array} \right] + \langle |S_{HV}|^2 \rangle \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$+ \begin{bmatrix} 2\text{Re}\langle S_{HH} S_{HV}^* \rangle & \langle S_{HH} S_{HV}^* \rangle + \langle S_{HV} S_{VV}^* \rangle \\ \langle S_{HV} S_{HH}^* \rangle + \langle S_{VV} S_{HV}^* \rangle & 2\text{Re}\langle S_{VV} S_{HV}^* \rangle \end{bmatrix}$$

Quad-pol data can be represented in two forms. One is the polarization scattering matrix S_2 , and the other is the polarization covariance matrix C_3 . Methods have already been developed to use CNN to recover the quad-pol C_3 matrix from the CP covariance matrix. However, no methods have yet been developed to restore quad-pol information from the S_2 . In quad-pol SAR data, the quad-pol covariance matrix C_3 is (3):

$$C_3 = \langle \underline{\Omega} \cdot \underline{\Omega}^* \rangle = \begin{bmatrix} \langle |S_{HH}|^2 \rangle & \sqrt{2}\langle S_{HH} S_{HV}^* \rangle & \langle S_{HH} S_{VV}^* \rangle \\ \sqrt{2}\langle S_{HV} S_{HH}^* \rangle & 2\langle |S_{HV}|^2 \rangle & \sqrt{2}\langle S_{HV} S_{VV}^* \rangle \\ \langle S_{VV} S_{HH}^* \rangle & \sqrt{2}\langle S_{VV} S_{HV}^* \rangle & \langle |S_{VV}|^2 \rangle \end{bmatrix} \quad (3)$$

The scattering matrix S_2 for quad-pol SLC SAR data is (4). There are two approaches to recovering quad-pol from compact polarization. The first is to restore the quad-pol scattering matrix from the target vector of CP, and the second is to restore the quad-pol covariance matrix C_3 from C_2 .

$$S_2 = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \quad (4)$$

A. Recover S_2 from Ω

This method deduces five unknowns in the quad-pol scattering matrix S_2 from four known elements in the target vector Ω . The CNN utilizes the real and imaginary parts of the target vector as input features to recover S_2 . To address the independence of these parts, two networks are employed—one for the real part and another for the imaginary part of the cross-polarization term S_{HV} , serving as labels for training and prediction. The CNN calculates these components as shown in Fig 1. Reciprocity ensures $S_{HV} = S_{VH}$ in the case of single-station backscatter. Substituting the predicted real and imaginary parts into Formula (1) allows the derivation of S_{HH} and S_{VV} , leading to the reconstruction of the quad-pol S_2 . The method for reconstructing the full-polarization scattering matrix S_2 does not make assumptions and relies directly on the excellent mapping capability of deep learning to fit and solve inverse problems. However, the stability of this direct, data-driven approach without

assumptions warrants further analysis.

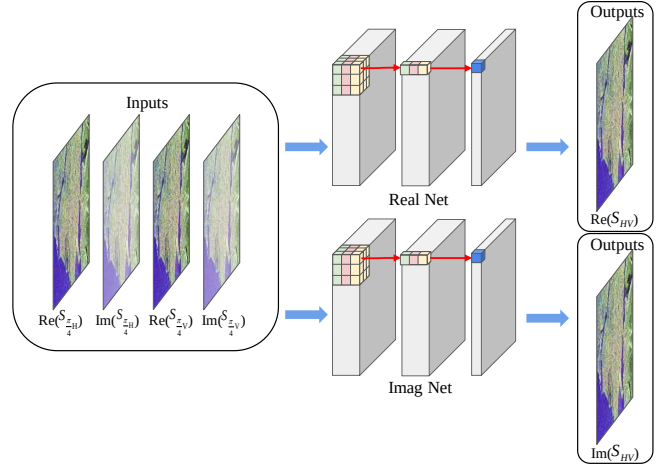


Fig. 1. CNN networks for S_2 Recover

B. Recover C_3 from C_2

The reflection symmetry is shown as (5). Under the condition of reflection symmetry, the compact polarimetric covariance matrix is represented as (6).

$$\langle S_{HH} S_{HV}^* \rangle \approx \langle S_{VV} S_{HV}^* \rangle \approx 0 \quad (5)$$

$$C_{3\text{recover}} = \begin{bmatrix} 2\langle J_{11} \rangle - \langle |S_{HV}|^2 \rangle & 0 & 2\langle J_{12} \rangle - \langle |S_{HV}|^2 \rangle \\ 0 & 2\langle |S_{HV}|^2 \rangle & 0 \\ 2\langle J_{12}^* \rangle - \langle |S_{HV}|^2 \rangle & 0 & 2\langle J_{22} \rangle - \langle |S_{HV}|^2 \rangle \end{bmatrix} \quad (6)$$

In formula (6), $\langle |S_{HV}|^2 \rangle$ represents the recovered cross-polarization term. Therefore, by eliminating 4 of the 9 unknowns in C_3 , the problem is transformed into solving for 5 unknowns using the 4 known values in C_{CP} . In the network for recovering quad-pol using the covariance matrix, the input features utilize the elements from C_2 , which include the real and imaginary parts and the modulus values of elements J_{11} , J_{22} and J_{12} . As known from formula (6), only the knowledge of $\langle |S_{HV}|^2 \rangle$ is required to obtain the restored pseudo covariance matrix. Therefore, $\langle |S_{HV}|^2 \rangle$ is used as the label for network training and prediction, as shown in Fig 2. $\text{Mod}(\cdot)$ represents the modulus operation.

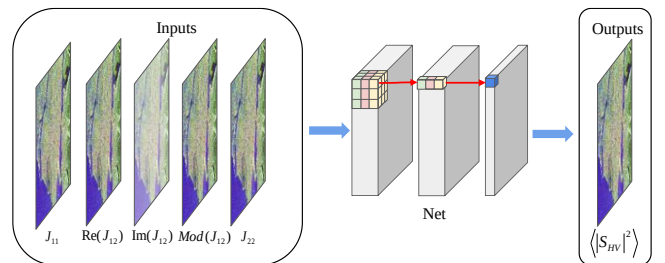


Fig. 2. C_2 Recover CNN network for C_3

C. The CNN structure

The CNN structure is shown in the Fig 3. First, a dual-branch CNN structure was constructed for pixel-level

recovery of compact polarized images, one branch being the polarization information extraction channel, primarily for perceiving surrounding pixel polarization information and extracting polarization features. The other branch is channel adaptive convolution, mainly for preserving the polarization information of each pixel while adapting to the input the channels of data.

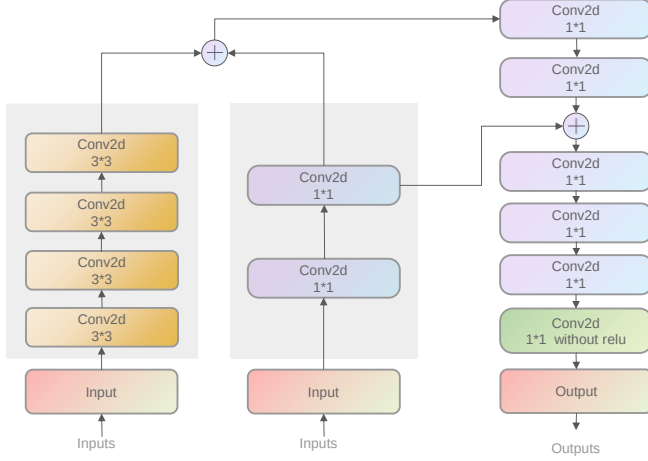


Fig. 3. CNN network structure diagram

3. EXPERIMENTAL RESULTS

A. Recover S_2 from Ω

This paper utilized 8 images from RadarSAT-2 and Gaofen-3 for model training and testing. Among these, 6 images were used for training and validation, while 2 images were used for model testing. Due to the large size of the images, during data preprocessing, the images were divided into 128×128 pixels for training and testing. The tested images were then reassembled back into their original image positions for recovering the entire image, which was subsequently evaluated. After data preprocessing, the training set comprised a total of 14,212 images, and the test set contained 8,448 images.

After model training through the CNN network structure, the images restored on the test set are as shown in the Fig 4. It can be observed that the overall restoration effect of the images is quite good, with details well-preserved and visually the results almost replicate the original images. There are some deviations in areas with partial grass cover over bare soil and in building areas, because densely built-up urban areas have secondary scattering and larger backscatter coefficients, resulting in some significant outlier values. During the CNN training process, less learning and fitting to this feature was performed. Grassland over bare soil exhibits volume scattering, which introduces certain random factors affecting the reflection of the radar beams.

The relative error heatmap generated for the cross-polarization term S_{HV} is shown in the Fig 5. The relative error is calculated by taking the bmp file for the cross-polarization term, which is the ratio of the difference between the

predicted value and the true value to the true value. It can be seen that most of the results have a difference of less than 0.2, with some areas showing a difference of 0.3. There is a certain numerical deviation, but it performs well in most areas.

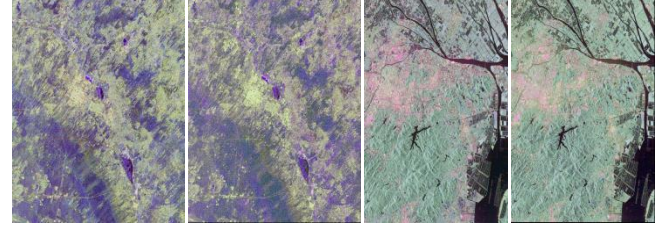


Fig. 4. Comparison of the original image and the Pauli RGB image generated from CNN prediction results. (a) (c) True value of Gaofen-3 and RadarSAT-2. (b) (d) Predicted value of Gaofen-3 and RadarSAT-2.

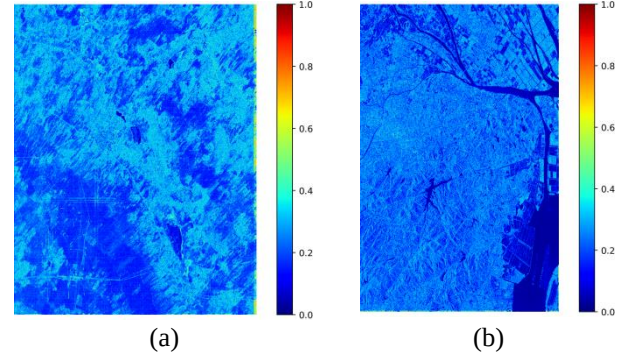


Fig. 5. Heat map of relative error between true and predicted values. (a) Gaofen-3 (b) RadarSAT-2

The probability estimation scatter plots between the restoration results and the true values for the cross-polarization term S_{HV} are shown in the Fig 6.

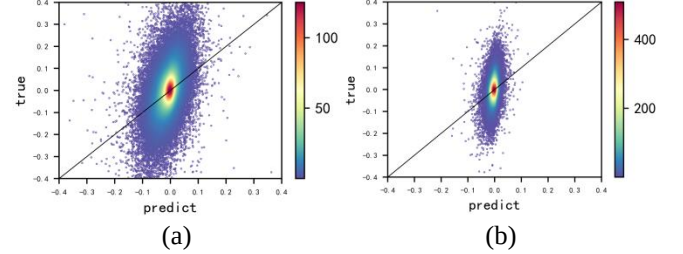


Fig. 6. scatter diagram (a) Gaofen-3 (b) RadarSAT-2

The area where the values are equal has the highest density of scatter points, and the areas where the predicted values differ significantly from the true values have a lower probability of scatter points. The overall distribution of the points with the highest probability density between the predicted values and the true values is along the line $y = x$. Meanwhile, the range of predicted values is smaller than that of the true values. This is because some random noise present in the true values is not removed in single-view data. Moreover, the data values in the true values are generally very small, so the fitting process did not effectively learn the distribution of the rarely occurring larger and smaller values.

B. Recover C_3 from C_2

AIRSAR is utilized to simulate the compact polarimetric covariance matrix C_2 . There are 27.12% pixels of SAN_FRANCISCO and 19.09% pixels of DEATHVALLEY are used for training, while the remaining pixel values are reserved for testing.

Pauli RGB images are shown in Fig 7. In the SAN_FRANCISCO, the overall restoration effect is relatively consistent with the real polarization covariance matrix, with the water being almost identically restored and good preservation of details. However, the vegetation in the mountain and farmland areas is less restored. In the DEATHVALLEY, the overall restoration is basically accurate, maintaining visual consistency with the original image. Apart from missing some water details, the building area shows some anomalous values due to the presence of double scattering.

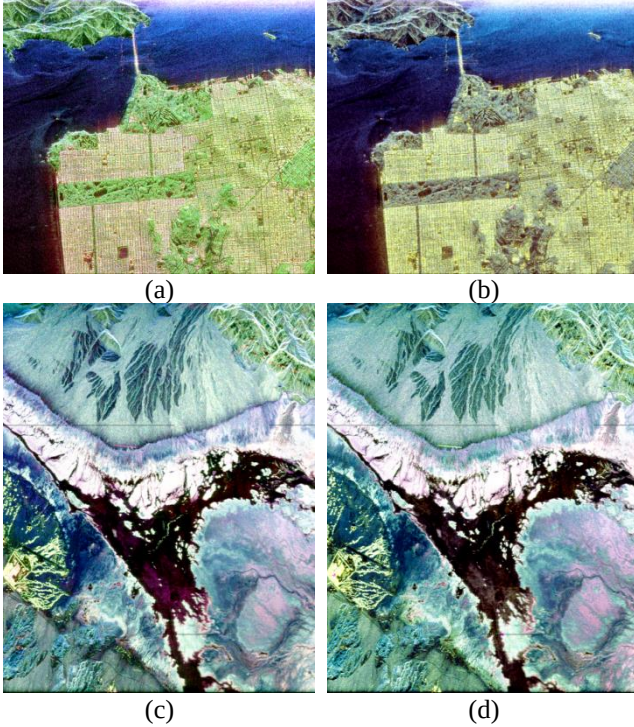


Fig. 7. Pauli RGB. (a) True value of SAN_FRANCISCO (b) Predicted value of SAN_FRANCISCO (c) True value of DEATHVALLEY (d) Predicted value of DEATHVALLEY

The predicted $\langle |S_{HV}|^2 \rangle$ are used to represent the ordinate in scatter plot Fig. 8, which is transformed from single-look data in Fig. 6 to multi-look C_3 matrix, altering its distribution and statistical properties. In DEATHVALLEY, not only is the overall probability distribution concentrated on the function $y = x$, but the overall distribution is also denser and closer to the true values, indicating higher accuracy. In SAN_FRANCISCO, the overall probability distribution is relatively concentrated, and the points with significant differences between the true values and the predicted values generally have a low probability, appearing in blue.

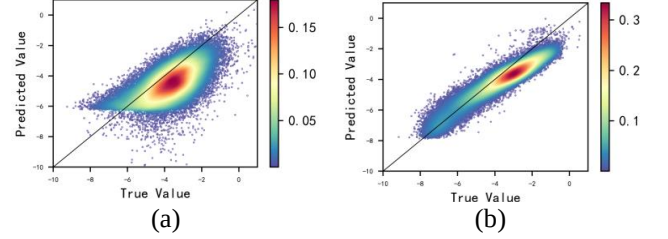


Fig. 8. Scatter plot. (a) SAN_FRANCISCO (b) DEATHVALLEY

C. Comparison of the accuracy of the two methods

TABLE 1. The MAE of two methods

METHOD	DATA	MAE
Ω to S_2	radarsat2	0.07626
	Gaofen-3	0.03824
C_2 to C_3	SAN_FRANCISCO	0.02703
	DEATHVALLEY	0.03493

MAE (Mean Absolute Error) are parameters that measure the error between predicted values and true values. As can be seen from the Table 1, both the Ω to S_2 and the C_2 to C_3 method perform well in terms of accuracy, with the method of C_2 to C_3 restoring images more accurately. The restoration effects of both CNN network methods are also feasible in terms of quantitative indicators.

Experiments showed that under the SLC condition, the accuracy of quad-pol images restored by the scattering matrix is 0.07626 and 0.03824. Restoration accuracy performs accurate with the same order of magnitude as that of the polarization covariance matrix. However, due to the lack of multi-view processing, it was impossible to eliminate the influence of random factors in ground target observation.

The polarization covariance matrix restoration method, due to the introduction of the prior assumption of reflection symmetry, has a certain applicable range in restoring quad-pol images of ground objects and does not have as strong a generalization capability as the Ω to S_2 method. At the same time, this method sacrifices some spatial resolution, and its visual interpretation effect is not as good as that of the S_2 matrix restoration. However, it performs better in terms of accuracy. By training and testing with data from airborne, spaceborne, and various sensors, and conducting tests in different regions, the results obtained ensure a certain level of generalizability in the model.

4. CONCLUSION

Two approaches for extracting polarization information initially through dual-branch CNN are introduced. One demonstrates broad applicability, qualitatively restoring and incorporating relaxed assumption constraints will be considered in future. While the other excels in accuracy metrics but requires a prior assumption of reflection symmetry, limiting effectiveness on certain ground objects.

5. REFERENCES

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